



# **Spaceship Titanic Project Report**

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# 1. Overview

This project uses supervised machine learning to predict whether passengers aboard the Spaceship Titanic were transported to another dimension after a spacetime anomaly. We worked with the Kaggle dataset, which provides passenger demographics, travel information, and luxury spending data. Our objective is to develop a predictive model that utilizes these features to accurately classify passenger outcomes.

## 2. Team Responsibilities

### 2.1 Team Lead – Arman Mahmoodzadeh

Coordinated work, assigned tasks, and ensured progress.

### 2.2 Pair A — Logistic Regression Team (Arman & Daniel)

- Performed categorical encoding
- Scaled numerical features
- Built a baseline Logistic Regression model
- Tuned hyperparameters
- Analyzed model coefficients
- Generated ROC and AUC metrics

### 2.3 Pair B — Decision Tree Team (Leo & Jose)

- Prepared data for tree-based models
- Tested different tree depths and visualized tree structures
- Analyzed which features appeared in the top splits
- Evaluated decision-tree accuracy, ROC, and AUC

### 2.4 All Members — Random Forest Model

- Combined the best preprocessing steps
- Trained Random Forest with multiple hyperparameter settings
- Analyzed feature importance
- Compared performance against baseline models

## 3. Dataset Descriptions

The Spaceship Titanic dataset from Kaggle consists of two files:

**train.csv (~8700 passengers)**

- Includes both features and the target label Transported.
- Used for model training and internal model evaluation.

#### **test.csv (≈4300 passengers)**

- Contains the same features but no target label.
- Used only for final prediction submission.

#### **Target Variable**

- Transported
  - True → Passenger was transported to another dimension
  - False → Passenger remained in normal space

### **3.1 Key Features**

| Feature                                               | Description                                                  |
|-------------------------------------------------------|--------------------------------------------------------------|
| HomePlanet                                            | Passenger's origin planet                                    |
| CryoSleep                                             | Whether the passenger was in cryogenic suspension            |
| Cabin                                                 | Deck/number/side location                                    |
| Destination                                           | Final intended planet                                        |
| Age                                                   | Age in years                                                 |
| VIP                                                   | Whether they purchased VIP services                          |
| RoomService / FoodCourt / ShoppingMall / Spa / VRDeck | Spending amounts                                             |
| Transported                                           | Target Variable: True (Transported), False (Not Transported) |

## **4. Tools & Environment**

All work was performed using:

- Python 3.x

- Jupyter Notebook
- NumPy, pandas
- matplotlib & seaborn
- scikit-learn (LogisticRegression, DecisionTreeClassifier, RandomForestClassifier)
- Graphviz for decision-tree visualization

## 5. Methods & Algorithms

### 5.1 Data Processing

- Split Cabin into Deck, Number, and Side components
- Identified categorical vs. numerical features
- Filled in missing numerical values with the median
- Filled in missing categorical values with the mode
- Removed irrelevant columns (Name, PassengerId, Cabin)
- One-hot encoded categorical features
- Split data into 80% training / 20% validation

### 5.2 Models Used

We used three supervised ML algorithms:

1. Logistic Regression
  - Scales numeric variables
  - One-hot encodes categorical variables
  - Provides a strong linear baseline
2. Decision Tree Classifier
  - Easy to visualize
  - Captures nonlinear patterns
  - Susceptible to overfitting
3. Random Forest Classifier
  - Reduces overfitting
  - Uses ensemble averaging
  - Higher accuracy and AUC

## 6. Model Performance

### 6.1 Logistic Regression

- Accuracy: 0.7918
- AUC: 0.8815
- Excellent baseline

- High separability between classes

## 6.2 Decision Tree

- Accuracy: 0.784
- AUC: 0.859
- Slightly lower than Logistic Regression
- Overfitting observed at depths  $> 7$

## 6.3 Random Forest

- Accuracy: 0.8085
- AUC: 0.8931
- Best overall model
- Shows improved generalization

# 7. Cross-Validation Results

These results reinforce that the Random Forest model is the strongest for this problem.

| Model               | Cross-Validation Accuracy |
|---------------------|---------------------------|
| Logistic Regression | 0.7872                    |
| Decision Tree       | 0.759                     |
| Random Forest       | 0.7851                    |

# 8. Model Interpretations

## 8.1 Logistic Regression Interpretation

- Strong baseline
- Sensitive to scaling and encoding
- AUC 0.881 indicates excellent separation

## 8.2 Decision Tree Interpretation

- Easy to interpret
- Captures nonlinearity
- More affected by missing values
- Lower accuracy compared to ensemble methods

## 8.3 Random Forest Interpretation

- Highest accuracy and AUC

- Reduces variance by averaging many trees
- Feature importance highlights CryoSleep, Cabin Deck, and HomePlanet

## 9. Final Conclusions

### 9.1 Best Model: Random Forest

- Achieved accuracy = 0.809 and AUC = 0.893
- Demonstrated the best generalization performance
- Offers better stability than a single Decision Tree
- Ideal candidate for final prediction submission

### 9.2 Key Insights

- CryoSleep has consistently been shown as the strongest predictor
- Cabin Deck and HomePlanet also carried significant weight
- Spending features (FoodCourt, VRDeck, etc.) were less impactful
- Data cleaning and processing had a major effect on the Logistic Regression performance

#### Results:

Transported Passengers: 2024 out of 4277

### 9.3 Project Summary

This project successfully applied three machine learning algorithms to a real-world scenario-based dataset. Based on both ROC/AUC analysis and cross-validation, the Random Forest model proved to be the most accurate and reliable for this problem. The project successfully met all its objectives and demonstrated the importance of thorough processing, model comparison, and evaluation.

## 10. References

### Kaggle Dataset

Kaggle. (2022). *Spaceship Titanic* [Data set].  
<https://www.kaggle.com/competitions/spaceship-titanic/data>

### scikit-learn Documentation

scikit-learn. (n.d.). *scikit-learn documentation*. <https://scikit-learn.org/stable/>